

VSG-100 Secure Network Gateway — Installation Guide

Technical writing sample by Nathan B. Smith. This document contains no export-controlled, CUI, or classified information. The equipment described is representative of real systems but is entirely fictional.

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Disclaimer

This guide is a writing sample and doesn't describe a real product.

Nathan B. Smith wrote this Installation Guide solely to demonstrate technical writing skill in hardware installation documentation. It isn't associated with any actual product or vendor.

- **Fictional company:** TechCoaches Information Systems Design is a fictional firm created for this and related writing samples.
- **Fictional equipment:** The VSG-100 Secure Network Gateway is a fictional system. It doesn't represent any real product, platform, or vendor architecture.
- **No export-controlled content:** This guide contains no technical data subject to the International Traffic in Arms Regulations (ITAR) or the Export Administration Regulations (EAR).

Content in this guide should be evaluated only as a demonstration of DITA authoring structure and hardware installation writing conventions.

About this guide

Scope, audience, and organization of this installation guide.

This guide describes how to unpack, mount, connect, and power on the VSG-100 Secure Network Gateway. Use it to bring a new VSG-100 unit into service, before you perform any security configuration.

This guide is for personnel who install and maintain network equipment in a secured facility. It assumes you're familiar with standard rack-mount hardware and basic networking cabling, but it doesn't assume prior experience with the VSG-100 specifically.

This guide covers:

- Site and environmental requirements
- Unpacking and inspecting the unit
- Mounting the VSG-100 in an equipment rack
- Connecting power and network cabling
- Powering on and verifying basic operation

This guide doesn't cover security configuration. After you complete installation, refer to the VSG-100 System Security Plan for procedures covering access control and boundary protection configuration.

Site and environmental requirements

Rack space, power, and environmental conditions required before you install the VSG-100.

Requirement	Specification
Rack space	1U, standard 19-inch equipment rack
Clearance	3 in (7.6 cm) minimum at the rear of the unit, to allow airflow through the rear-panel cooling fan exhaust vents and access for cabling
Power	100-240 VAC, 50-60 Hz, dedicated circuit recommended

Requirement	Specification
Operating temperature	32°F to 104°F (0°C to 40°C)
Operating humidity	10% to 90%, noncondensing
Network access	One available port on the tactical network segment and one available port on the program office network, as described in the VSG-100 System Security Plan

Important: Don't share the VSG-100's power circuit with high-draw equipment, such as space heaters or other rack units with large startup current draw. Shared circuits increase the risk of nuisance trips.



Note: Values in this table are fictional and provided to demonstrate reference-topic conventions. See [Disclaimer](#) on page 3 for the full disclaimer.

Unpack and inspect the VSG-100

Unpack the VSG-100 and confirm the shipment is complete and undamaged before you install it.

Before you begin, clear a flat, static-safe work surface near the installation site.



CAUTION: The VSG-100 contains components sensitive to electrostatic discharge (ESD). Wear an ESD wrist strap, and keep the unit on an antistatic mat until it's installed in the rack.

1. Inspect the shipping carton for visible damage before opening it.

If the carton is damaged, photograph it before opening, and contact your shipping provider.

2. Open the carton, and remove the packing materials.



Note: Retain all packaging until the unit passes verification in [Power on and verify the VSG-100](#) on page 6. You'll need the original packaging for any warranty return.

3. Confirm the shipment contains the following items:

- One VSG-100 Secure Network Gateway unit
- One AC power cord
- One rack-mount kit (two rails, mounting screws)
- Quick start card

4. Inspect the unit for physical damage, including the chassis, connectors, and mounting ears.

Restriction: Don't power on a unit that shows signs of physical damage. Contact your equipment supplier instead.

5. Record the unit's serial number, printed on the underside of the chassis, in your asset inventory.

You've confirmed the shipment is complete and the unit is undamaged, and you're ready to prepare the installation site.

Mount the VSG-100 in an equipment rack

Install the VSG-100 in a standard 19-inch equipment rack using the supplied rack-mount kit.

- Confirm the site meets the requirements in [Site and environmental requirements](#) on page 3
- Rack-mount kit (supplied)
- Phillips screwdriver
- A second person to help support the unit's weight during installation



Warning: Confirm the rack is secured and stable, with anti-tip supports engaged if the rack isn't permanently fixed to the floor. A rack that shifts or tips during installation can cause serious injury.



CAUTION: The VSG-100 requires two people to lift safely. Lifting it alone risks personal injury and can drop and damage the unit.

1. Attach a mounting rail to each side of the rack at the height you've selected for the VSG-100.
Use two mounting screws per rail, at minimum.
2. With a second person supporting the opposite side, lift the VSG-100 into place on the rails.



CAUTION: Keep fingers clear of the gap between the chassis and the rail while positioning the unit, to avoid pinch injury.

3. Align the mounting ears with the rack's mounting holes.
4. Secure the unit to the rack using the supplied mounting screws, two per side.
5. If your rack is a four-post rack, attach a rear support bracket to the VSG-100's rear support bracket mounting holes.
See [VSG-100 rear panel](#) on page 8 to locate these mounting holes. A rear support bracket isn't required on a two-post rack.
6. Confirm the unit is level and firmly seated before releasing support.

The VSG-100 is mounted and ready for power and network connections.

Connect power to the VSG-100

Connect the VSG-100 to a power source without powering it on.

- The VSG-100 mounted in the rack (see [Mount the VSG-100 in an equipment rack](#) on page 4)
- Supplied AC power cord
- A dedicated, grounded power circuit



Warning: Electric shock hazard. Confirm the power circuit is properly grounded, and don't connect the VSG-100 to power with wet hands or in a wet environment.



CAUTION: Confirm the power source matches the VSG-100's rated input (100-240 VAC, 50-60 Hz) before connecting. Connecting an unrated power source can damage the unit.

1. Confirm the VSG-100's power switch, if equipped, is in the off position.
2. Connect an approved grounding strap from the rack's grounding bus to the chassis ground lug on the VSG-100 rear panel.

See [VSG-100 rear panel](#) on page 8 to identify the ground lug.

3. Connect the AC power cord to the power connector on the VSG-100 front panel.

See [VSG-100 front panel](#) on page 7 to identify the power connector.

4. Route the power cord along the rack's cable management arm, away from network cabling.



CAUTION: Don't block the cooling fan exhaust vents on the VSG-100 rear panel with cabling. Restricted airflow can cause the unit to overheat.

5. Connect the other end of the power cord to the dedicated power circuit.
6. Don't power on the unit yet.

Important: Complete [Connect network interfaces on the VSG-100](#) on page 6 before you power on the VSG-100. Powering on before network connections are in place can cause the unit to come up in an unexpected state.

The VSG-100 has power available, and you're ready to connect network cabling.

Connect network interfaces on the VSG-100

Connect the VSG-100's Interface A and Interface B ports to the tactical network segment and the program office network.

- The VSG-100 mounted and connected to power (not yet powered on)
- Two shielded network cables rated for your network's speed
- Confirmed port assignments on the tactical network segment and the program office network

Interface A connects to the tactical network segment. Interface B connects to the program office network. See [VSG-100 front panel](#) on page 7 to locate both ports.

Important: Connecting a cable to the wrong interface can expose CUI to an unauthorized network segment. Confirm each cable's origin against your approved network diagram before you connect it.



CAUTION: The network connectors are sensitive to electrostatic discharge. Continue to observe ESD precautions while making these connections.

1. Connect one network cable from the tactical network segment to Interface A on the VSG-100 front panel.
2. Connect a second network cable from the program office network to Interface B on the VSG-100 front panel.
3. Route both cables along the rack's cable management arm, and label each cable at both ends.
Consistent labeling makes later maintenance and troubleshooting faster.
4. Confirm both cable connections are fully seated.



Note: A cable that isn't fully seated can cause an intermittent link, which is difficult to diagnose after the unit is in service.

The VSG-100 is connected to both networks, and you're ready to power on the unit.

Power on and verify the VSG-100

Power on the VSG-100, and confirm it starts up correctly before you begin security configuration.

- The VSG-100 mounted, powered, and connected to both networks
- Completed [Connect network interfaces on the VSG-100](#) on page 6



Warning: If you notice smoke, an unusual smell, or an unusual sound while powering on the VSG-100, disconnect power immediately, and don't reconnect it. Contact your equipment supplier before further use.

1. Turn on the VSG-100 using the front-panel power switch, if equipped, or by confirming power is applied.
2. Watch the **PWR** LED. Confirm it turns solid green within 10 seconds.
If the **PWR** LED doesn't turn on, confirm the power connection from [Connect power to the VSG-100](#) on page 5.
3. Watch the **SYS** LED during startup. Confirm it changes from blinking to solid green within approximately 90 seconds.
A solid green **SYS** LED indicates the unit has completed startup and is ready for configuration.
4. Confirm the **ALM** LED is off.



CAUTION: If the **ALM** LED is lit, don't proceed to security configuration. Refer to the VSG-100 maintenance manual for alarm troubleshooting.

5. Confirm link activity on both Interface A and Interface B.

Each interface should show a steady link indication once its connected network segment is active.

The VSG-100 has started successfully and is ready for initial configuration. Refer to the VSG-100 System Security Plan for the access control and boundary protection procedures that follow installation.

VSG-100 front panel

Physical layout and connector identification for the VSG-100 front panel.

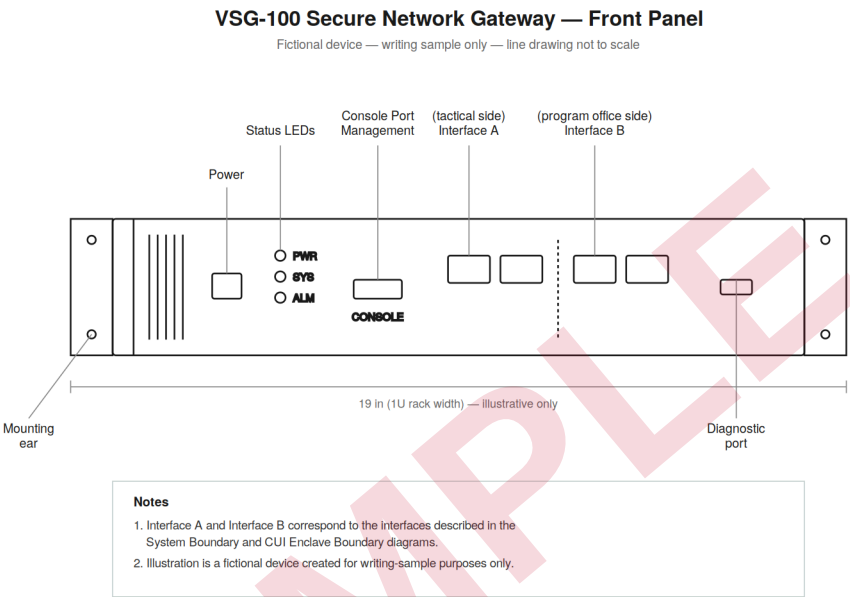


Figure 1: VSG-100 front panel

Item	Description
Power	Power connector. Confirm the unit is powered down before connecting or disconnecting.
Status LEDs	PWR , SYS , and ALM indicators. See Power on and verify the VSG-100 on page 6 for indicator states during power-on.
Management console port	Administrator access point used to reach the VSG-100 management console after installation is complete.
Interface A	Tactical-side network interface. Connect during Connect network interfaces on the VSG-100 on page 6.
Interface B	Program-office-side network interface. Connect during Connect network interfaces on the VSG-100 on page 6.
Diagnostic port	Local diagnostic access only. Don't use this port to pass production network traffic.
Mounting ears	Standard 19-inch, 1U rack-mount hardware. Used during Mount the VSG-100 in an equipment rack on page 4.



Note: This illustration is fictional. See [Disclaimer](#) on page 3 for the full disclaimer. For the ground lug, cooling vents, and other rear-facing connectors, see [VSG-100 rear panel](#) on page 8.

VSG-100 rear panel

Physical layout and connector identification for the VSG-100 rear panel.

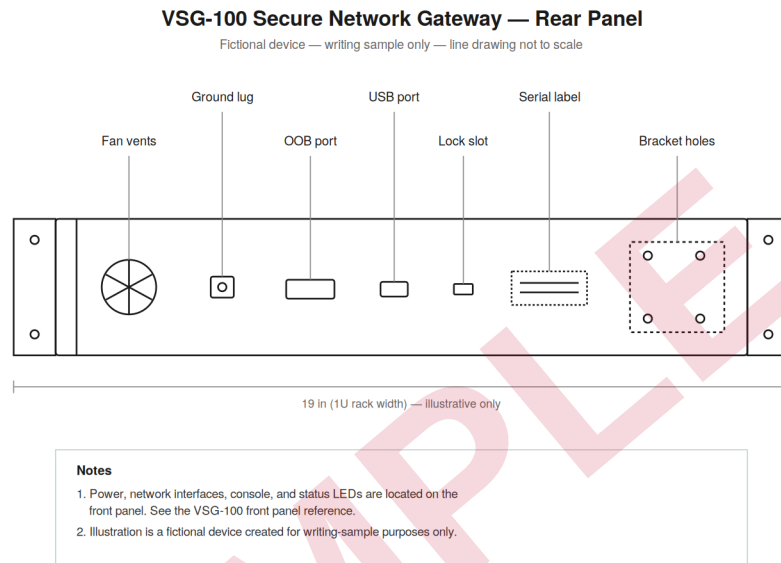


Figure 2: VSG-100 rear panel

Item	Description
Cooling fan exhaust vents	Don't block these vents. See Connect power to the VSG-100 on page 5 for cable-routing guidance that keeps this airflow path clear.
Chassis ground lug	Connection point for the equipment grounding conductor. Connected during Connect power to the VSG-100 on page 5.
OOB management port	Secondary, out-of-band administrative access point, separate from the front-panel management console port. Reserved for future use in this guide.
USB recovery port	Used for firmware recovery only. Refer to the VSG-100 maintenance manual before using this port.
Security lock slot	Accepts a standard cable lock to help deter unauthorized physical removal of the unit.
Asset / serial number label	Confirm this matches the serial number you recorded during Unpack and inspect the VSG-100 on page 4.
Rear support bracket mounting holes	Used with a four-post rack's rear support bracket, if your rack requires one. See Mount the VSG-100 in an equipment rack on page 4.



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